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The Editors

Special Issue on *Computer Vision Systems for Document Analysis and Recognition*
International Journal on Document Analysis and Recognition (IJDAR)

Dear Editors,

We are pleased to submit our manuscript, “**Reference-Conditioned Structural Assessment of Low-Resource Chinese Calligraphy with Overlapping Stroke Parsing and Human-Audited Spatial Scoring**”, by Xiaofan Liu, Ronghao Zhang and Yuan Feng (Software Engineering Institute, East China Normal University), for consideration in the above special issue. Xiaofan Liu is the corresponding author.

Fit to the special issue. The special issue invites work that goes beyond narrowly defined benchmarks and addresses document heterogeneity, low-resource learning, and explainable evidence. Our manuscript studies a document-analysis problem that is deliberately difficult on all three axes: the inputs are brush-based Chinese calligraphy, the structural annotation is scarce (769 quality-controlled samples over 40 character identities, without reliable writer identities), and the intended output is not a single number but inspectable evidence. We formulate *reference-conditioned structural assessment*—comparing a learner instance against an approved reference of the same character—and report where the approach works, where it does not, and under which conditions each claim holds.

Contributions. The manuscript is organized around four research questions.

- *RQ1.* We define an overlapping six-channel representation (five direction families that may overlap, plus a separate endpoint channel) and compare U-Net, DeepLabV3+ and SegFormer-B2 across three seeds on both a standard and a character-disjoint split, showing that the best in-corpus parser is not the best parser for unseen characters.
- *RQ2.* Using controlled nuisance and structural perturbations of 200 real references, we quantify what a bounded similarity registration buys: nuisance penalties fall substantially relative to no alignment while structural edits remain penalized. We also report a wider-search comparator and a channel-coverage score audit.
- *RQ3.* Three blinded domain-qualified raters scored 150 natural same-character pairs. We report inter-rater reliability, the association of three scores, and a direct-ink control that isolates how much the scalar score depends on the parser. Retrospective development is explicitly separated from character-grouped internal validation.
- *RQ4.* We evaluate localized diagnostic evidence under interventions with known ground truth and publish a failure taxonomy rather than an aggregate accuracy.

All reported values are computed from frozen artifacts whose SHA-256 digests are listed in the Online Resource.

Originality. This submission is original work. It has not been published previously and is not under consideration elsewhere. It does not extend a prior conference or journal publication by the same authors, so there is no companion paper to declare or to supply for comparison.

Third-party material and permission. Three figures reproduce handwriting images from the Calli-Tongji open subset (Ouyang Xun regular-script and Wang Xizhi running-script categories), which is distributed under CC BY-NC 4.0. These images are used with full attribution, at reduced resolution, and only as illustrative examples; the dataset itself is not redistributed, and the manuscript states this under Data Availability. Because the licence is non-commercial while

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Format. The manuscript is 20 pages in the IJDAR two-column format, including references, figures and tables, and therefore within the journal’s 20-page guidance. It uses a structured abstract of 228 words (Purpose, Methods, Results, Conclusion) and six keywords. Complete editable LaTeX source, including the class file and all figures, is uploaded as a single archive named `OneStroke2026_manuscript_latex.zip`, in which `manuscript.tex` is the main document. The Electronic Supplementary Material (four tables and two notes) is uploaded separately as `OneStroke2026_ESM_1_submission.pdf` and is cited in the text as Online Resource 1.

Declarations. All authors have approved the submission and agree to its content. The authors declare no competing interests. The work received no external funding. The study involved no clinical, medical, or behavioural intervention; the handwriting corpus was commissioned without recording writer identities, and the similarity ratings were provided under evaluator codes by two of the authors and one collaborating specialist. Data and code availability statements are given in the manuscript and entered in the submission interface.

Reviewer suggestions. We respectfully suggest reviewers with expertise in document image analysis and in computational calligraphy, and we ask that experts from the authors’ institution be excluded.

We would be glad to provide any additional material the editors require. Thank you for considering our manuscript.

Sincerely,

Xiaofan Liu

on behalf of all authors

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